

Differential Equations & Vector Calculus

M-II | R23 Regulation | JNTUK Kakinada

Year / Sem : 1st Year – 2nd Semester (1-2)

Branch : Common for All Branches

Credits : 3 (Theory)

Category : Mathematics – II (M-II)

■ Course Objectives

- Learn differential equations for engineering problems
- Understand multivariable & vector calculus concepts
- Apply mathematics in real-world engineering systems

■ Course Outcomes

- Solve first & higher order differential equations
- Work with PDEs in physical & thermal systems
- Understand vector fields: gradient, divergence, curl
- Calculate work done, flux & circulation integrals

UNIT – I : First Order Differential Equations

Types of 1st Order Equations

- Variable separable method – direct integration approach
- Homogeneous equations – substitution $y = vx$
- Linear differential equations – integrating factor method
- Bernoulli's equation – reducible to linear form
- Exact differential equations – condition: $\partial M/\partial y = \partial N/\partial x$

Real-World Applications

- Newton's Law of Cooling – temperature-time relation
- Growth & Decay models – population, radioactive decay
- Electrical Circuits – RL circuit, charging & discharging

UNIT – II : Higher Order Differential Equations

Linear DEs with Constant Coefficients

- Homogeneous equations – auxiliary equation (AE) method
- Non-homogeneous equations – CF + PI structure
- Complementary Function (CF) – real, complex & repeated roots

Finding Particular Integral (PI)

- PI for e^{ax} , $\sin(ax)/\cos(ax)$, x^n , $e^{ax}V(x)$ using operators
- Wronskian – test for linear independence of solutions
- Method of Variation of Parameters – general PI method

Engineering Applications

- LCR Circuit – analogous to mechanical vibrations
- Simple Harmonic Motion – undamped, damped, forced vibrations

UNIT – III : Partial Differential Equations (PDE)

Formation of PDEs

- By elimination of arbitrary constants from $z = f(x,y)$
- By elimination of arbitrary functions – total differentiation

Solution Methods

- Lagrange's Method – $Pp + Qq = R$, auxiliary equations
- Charpit's method – non-linear first order PDEs
- Linear PDEs with constant coefficients – operator method

Standard Forms

- Standard Form I: $F(p, q) = 0$ – only p and q
- Standard Form II: $F(z, p, q) = 0$ – z , p and q
- Standard Form III & IV – separable and Clairaut forms

UNIT – IV : Vector Differentiation

Vector Functions & Fields

- Scalar point function vs Vector point function
- Limit, continuity and derivative of vector functions

Del Operator (∇) & Applications

- Gradient (∇f) – maximum rate of change, normal to surface
- Divergence ($\nabla \cdot F$) – outflow/source measure of a vector field
- Curl ($\nabla \times F$) – rotation/circulation of a vector field
- Directional derivative – rate of change in a given direction

Vector Identities

- $\nabla(fg)$, $\nabla \cdot (fF)$, $\nabla \times (fF)$, $\nabla \cdot (F \times G)$
- Solenoidal field: $\nabla \cdot F = 0$ | Irrotational field: $\nabla \times F = 0$
- Laplacian: $\nabla^2 f = \text{div}(\text{grad } f)$ and its properties

UNIT – V : Vector Integration

Line Integrals

- Work done by a force: $W = \int F \cdot dr$ along a curve C
- Circulation of a vector field around a closed path

Surface & Volume Integrals

- Surface integral – flux of F across surface S : $\int F \cdot n \, dS$
- Volume integral – $\int F \, dV$ over a region V

Integral Theorems

- Green's Theorem – line integral $\leftrightarrow$ double integral (2D)
- Stokes' Theorem – line integral $\leftrightarrow$ surface integral (3D curl)
- Gauss's Divergence Theorem – surface $\leftrightarrow$ volume integral

Engineering Applications

- Fluid flow – velocity fields, flux computation
- Electromagnetic fields – Faraday's & Gauss's law forms
- Work-energy calculations in mechanical systems

■ Key Formulas to Remember

- ◆ Integrating Factor: $e^{\int P dx}$ for linear DE
- ◆ CF roots: real, complex, repeated – 3 cases
- ◆ Lagrange: $Pp + Qq = R$ auxiliary $dx/P = dy/Q = dz/R$
- ◆ Gradient: $\nabla f = (\partial f/\partial x)\mathbf{i} + (\partial f/\partial y)\mathbf{j} + (\partial f/\partial z)\mathbf{k}$
- ◆ Divergence: $\nabla \cdot \mathbf{F} = \partial F_1/\partial x + \partial F_2/\partial y + \partial F_3/\partial z$
- ◆ Stokes: $\oint \mathbf{F} \cdot d\mathbf{r} = \iint (\nabla \times \mathbf{F}) \cdot \mathbf{n} dS$

■ Important Notes & Exam Tips

- ◆ Also known as M-II (Mathematics-2) for all branches
- ◆ Practice numericals daily – theory alone won't help
- ◆ High-weightage for GATE & university exams
- ◆ Learn all 3 integral theorems with conditions
- ◆ Derivations of PI methods carry exam marks
- ◆ Vector identities – learn proofs, not just results